Quantum Reservoir Computing for Forecasting Equity Volatility Regime Transitions

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Global Industry Challenge 2026 · Phase 3 Submission · Track A

1. Introduction and contribution

Volatility regimes influence risk limits, hedging and execution, while transitions are precisely the periods in which persistence assumptions can fail. We therefore separate two related tasks: classifying the low/medium/high regime of forward realised variance, including transition discrimination, and forecasting its continuous magnitude. Reservoir computing is attractive because a fixed nonlinear dynamical system can expose a rich feature map while training only a classical readout [1, 2].

We evaluate rather than assume the value of a compact quantum reservoir. The study combines leakage-safe, validation-only architecture selection; complete classical controls; controlled finite-shot/noise analysis; formal paired statistics; a genuine-MNIST benchmark; and independent qBraid clean-room reproduction. All quantum states are produced by exact classical simulation. The aim is a transparent account of when the selected QRC is competitive, not a claim of quantum advantage.

2. Data, targets and evaluation protocol

The frozen public snapshot `yahoo_chart_20100101_20251231_v1` contains adjusted daily observations for SPY and QQQ and levels for VIX from 4 January 2010 through 31 December 2025. Sixteen causal inputs cover returns, trailing realised and Parkinson volatility, volume/range, VIX, QQQ and weekday families. Preprocessing statistics are fitted on training data only.

Chronological splits contain 2,744 training observations (2 February 2010–23 December 2020), 748 validation observations (4 January 2021–21 December 2023), and 497 test observations (2 January 2024–23 December 2025). Five trading days are purged at each boundary, and no sample is shuffled.

For daily log return r_t, the target is the annualised forward five-day realised variance

$$y_t = \frac{252}{5}\sum_{k=1}^{5} r_{t+k}^2.$$

Low, medium and high labels use only the training-target 0.33 and 0.66 quantiles (0.00663709 and 0.01954608). A transition is a future label different from the current trailing-volatility label. Classification uses macro-F1, balanced accuracy and transition PR-AUC; continuous forecasts use QLIKE, RMSE, MAE and correlation. QLIKE is emphasised because it is robust for variance forecast evaluation [3].

3. Final QRC and classical controls

The final financial reservoir has two qubits, two virtual nodes and the `reset_each_input` policy. Each market observation starts from the configured initial density matrix; within that observation, the input qubit is partially traced and re-injected at each virtual substep, so state is retained across the two substeps but never across market dates. A fixed seeded Gaussian projection $b_v^\top x_t$ sets $\theta_{t,v} = 0.5 b_v^\top x_t$, encoded as $R_y(\theta_{t,v})|0\rangle$ on the input qubit.

Each substep evolves for half of total time $\tau = 1$ under the fixed random transverse-field Ising Hamiltonian

$$H = \sum_{(i,j)\in E} J_{ij} Z_i Z_j + \sum_i h_i X_i,$$

where E is the unique edge of the two-qubit ring, seeded $J_{ij} \sim U[-1,1]$ and $h_i \sim U[0.5, 1.5]$. Expectations of Z_0, Z_1, Z_0Z_1 at each virtual node give six features. This temporal multiplexing increases observable density without increasing qubit count; it does not create cross-date memory.

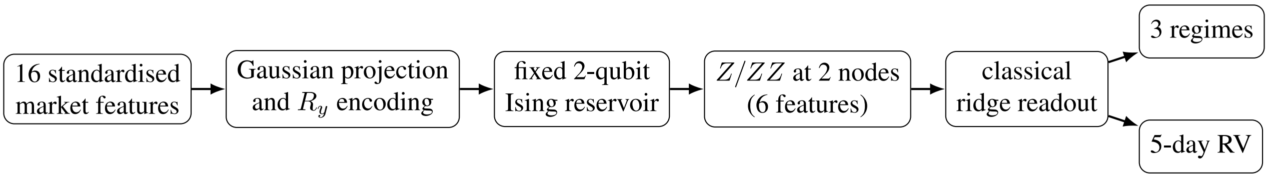

Figure 1: Final financial pipeline. The fixed exact-simulation reservoir creates six nonlinear observables per date; only the two classical readouts are fitted. Architecture and ridge choices use validation data, never test outcomes.

Unpenalised-intercept ridge readouts predict three one-hot regime indicators (then apply a stable softmax) and log realised variance (then exponentiate). Candidate ridge penalties $10^{-5}, 10^{-3}, 10^{-1}, 1$ are selected on validation macro-F1 or QLIKE. Results average the frozen reservoir seeds 2026, 2027 and 2028; probabilities and forecasts are not ensembled.

Strong controls test whether the QRC adds value rather than supply easy comparisons: majority and regime-persistence classifiers, logistic regression, realised-variance persistence, an Echo State Network (ESN), and a Gaussian GARCH(1,1) [4]. Every comparable model uses the same dates and target definitions.

4. Financial results and statistical analysis

Table 1: Frozen 2024–2025 financial test results. Higher is better except for QLIKE, RMSE and MAE. QRC is mean across three reservoir seeds; the full-precision population standard deviations are retained in the source manifest. Bold marks the best listed point estimate per metric. GARCH has no transition score.

Classifier	Macro-F1	Bal. acc.	Trans. PR-AUC
Majority classifier	0.151	0.333	0.506
Regime persistence	0.485	0.485	0.511
Logistic regression	**0.526**	**0.529**	**0.683**
ESN	0.446	0.478	0.645
GARCH(1,1)	0.373	0.442	—
QRC mean	0.454	0.491	0.640

Forecaster	QLIKE	RMSE	MAE	Corr.
RV persistence	-2.397	0.0866	0.0255	0.335
ESN	-2.865	**0.0662**	**0.0183**	0.532
GARCH(1,1)	**-2.895**	0.0740	0.0233	0.356
QRC mean	-2.814	0.0894	0.0216	**0.543**

The QRC is competitive in selected dimensions but does not dominate. Logistic regression leads regime classification (macro-F1 0.526 versus QRC 0.454); GARCH leads QLIKE (-2.895 versus -2.814); and ESN leads RMSE (0.0662 versus 0.0894) and MAE (0.0183 versus 0.0216). QRC attains the highest listed forecast correlation (0.543) and a transition PR-AUC of 0.640, but point-estimate leadership on one metric is not systematic superiority.

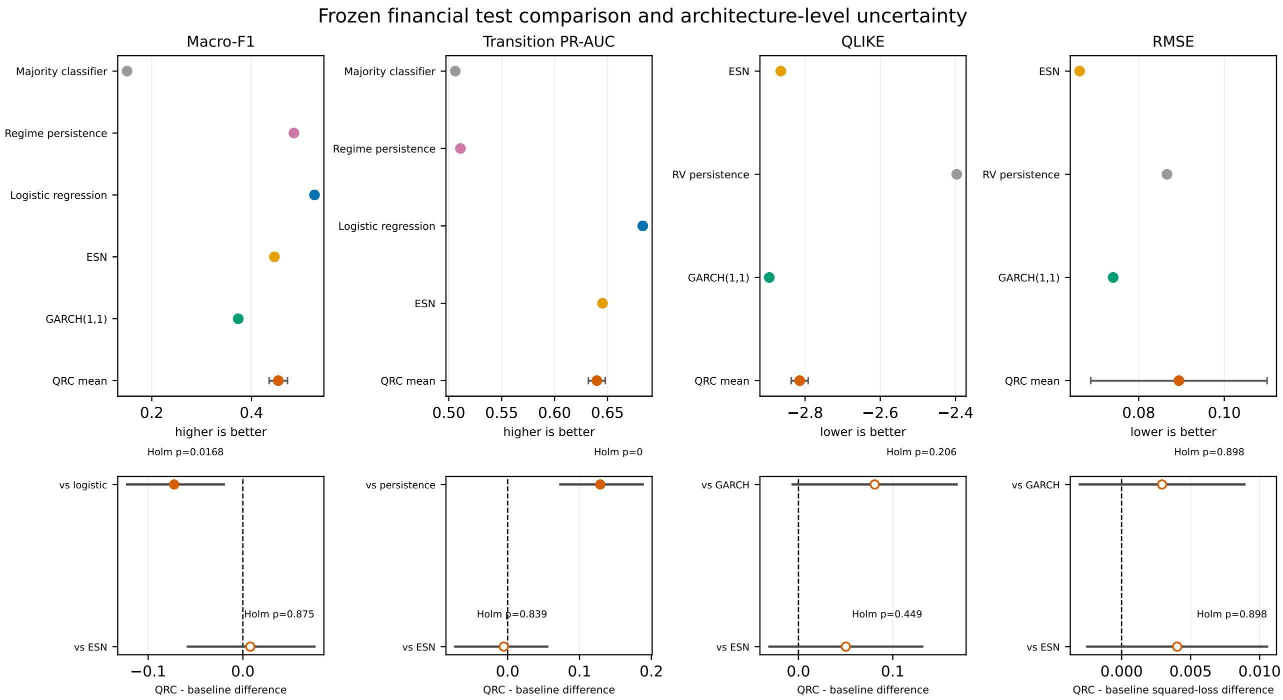

Figure 2: Frozen financial test estimates and selected architecture-level 95% intervals for QRC-minus-baseline differences. Filled markers denote Holm-adjusted support and hollow markers do not. QRC statistics average paired per-seed differences, not prediction ensembles. Logistic leads macro-F1, GARCH leads QLIKE and ESN leads RMSE.

Formal tests use 10,000 paired replicates. Financial losses use a Diebold–Mariano-style test [5] with Bartlett

Newey–West HAC lag 4 [6], a five-step finite-sample correction and circular-block intervals; classification metrics are recomputed within paired circular blocks. Holm correction is applied within predeclared metric families. QRC-minus-logistic macro-F1 is -0.0726 (95% interval $[-0.1223, -0.0202]$, adjusted $p = 0.0168$). QRC exceeds regime persistence in transition PR-AUC by 0.1289 (95% interval $[0.0733, 0.1880]$, adjusted resampling $p = 0$ at the 10,000-replicate resolution). No QRC-versus-ESN financial classification metric, nor any financial regression architecture comparison, is Holm-significant. Failure to reject is not evidence of equivalence.

Post-freeze diagnostics preserve training-derived regime and tail thresholds. They examine calibration, transition types, rolling errors and high-volatility tails without refitting. These checks expose limited probability calibration, small tail samples and a near-singular feature matrix for one seed; they are descriptive and do not alter the Stage 2A inference or model selection.

5. Genuine MNIST, robustness and portability

The independent MNIST benchmark uses checksum-pinned official IDX files [7] and balanced, disjoint splits of 6,000 training, 1,000 validation and 1,000 official-test images. Each 28×28 image is scaled by 255 and compressed row-wise into five fixed column-band means. The five-qubit, two-virtual-node ring reservoir resets between images and carries state only across the 28 rows of one image. Five Z and five ring-edge ZZ observables per virtual node are reduced by seven fixed temporal summaries to 140 image features. Training-only standardisation and validation macro-F1 select the multinomial-logistic readout.

Table 2: Genuine-MNIST test results. QRC values are mean $\pm$ population standard deviation across the three frozen reservoir seeds; baselines are single deterministic fits.

Model	Accuracy	Macro-F1	Bal. acc.	Macro AUC
Flattened logistic	0.889	0.888	0.889	0.987
Size-controlled ESN	**0.934**	**0.934**	**0.934**	**0.995**
Exact QRC mean	0.874 ± 0.005	0.874 ± 0.005	0.874 ± 0.005	0.987 ± 0.001

Frozen exact QRC mean accuracy is 0.874 ± 0.005, compared with 0.889 for flattened logistic regression and 0.934 for the size-controlled ESN. The ESN is Holm-significantly higher on all four reported metrics; QRC versus logistic is not Holm-significant. For seed 2026, accuracy falls from 0.871 analytically to 0.504 with 2,048 sampled measurements, 0.492 with added depolarising probability 0.01, and 0.485 with measurement-flip probability 0.02. These controlled simulations show finite-shot/noise sensitivity, not physical-device performance [8].

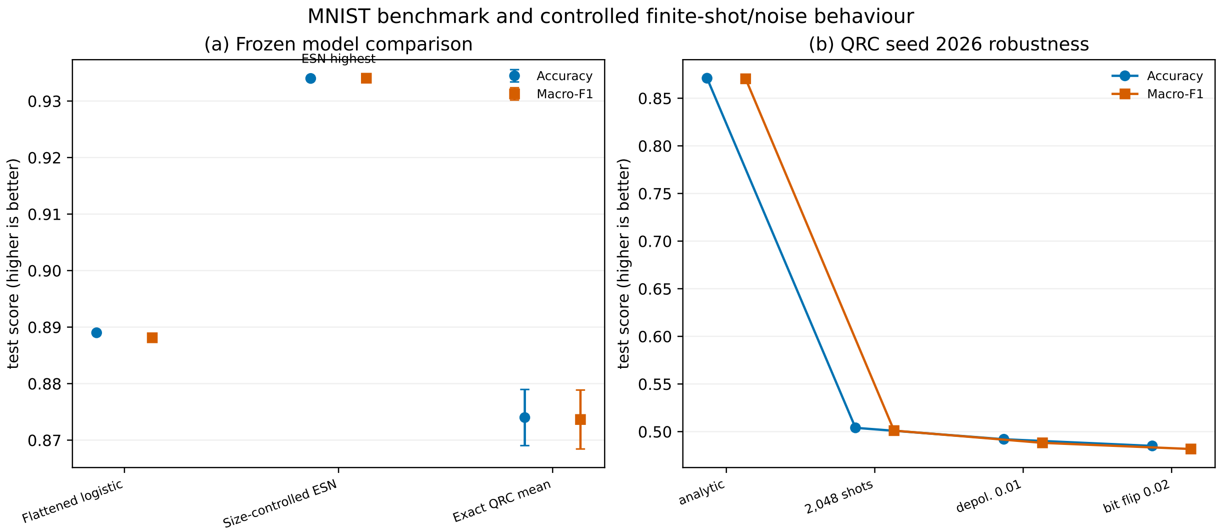

Figure 3: Genuine-MNIST test comparison and controlled robustness. Exact QRC bars show mean and population variation across three seeds; finite-shot/noise rows use frozen seed 2026. The 2,048-shot, depolarising and measurement-flip conditions are classical simulations, not hardware-calibrated experiments.

The qBraid e0l4 clean-room run (Python 3.12.3, Linux/x86_64) verified identical official files, partitions, preprocessing, architecture, seeds and protocol, yet reproduced QRC mean accuracy 0.875 rather than the frozen macOS/ARM 0.874. Across 3,000 test seed–image evaluations, 14 predictions changed, for a net gain of three correct predictions. Reservoir features first differ only at floating-point last bits (at most 6.44×10^{-15}) and, with either frozen readout, change no decisions. The divergence enters when platform-sensitive BLAS/L-BFGS convergence selects nearby multinomial-readout iterates on ill-conditioned designs. The accepted contract checks features, coefficients, scores, predictions and confusion matrices; rankings and scientific conclusions are unchanged.

6. Limitations, reproducibility and conclusion

The main limitations are classical exponential-cost state simulation, only three reservoir seeds, overlapping forward targets, small high-volatility tails, information loss in the MNIST band compression, and sharp degradation under the frozen finite-shot conditions. The controlled noise channels are not hardware calibrated, and no physical QPU was executed.

Reproduction uses a checksum-pinned Python environment and a resumable orchestrator. Immutable raw inputs remain SHA-256 exact. Generated financial tables require exact rows, dates, splits, labels, missingness and thresholds; finite decimals are canonically committed at 10 significant digits, whose maximum relative quantisation width is 10^{-9}. A separately gated GARCH portability contract accepts only equivalent optimiser endpoints and forecast paths; the observed metric drift was below 9×10^{-8}, inside a GARCH-only 2.5×10^{-7} metric envelope, with unchanged threshold crossings, rankings and displayed values. The qBraid full workflow also regenerated statistical, diagnostic and publication artefacts.

The compact QRC demonstrates useful nonlinear expressivity and competitive performance in selected settings, but it establishes neither quantum advantage nor systematic superiority over strong classical models. Its principal value here is a falsifiable, reproducible assessment of where a small exact-simulation QRC is competitive, where classical methods remain stronger, and how numerical portability affects hybrid quantum–classical workflows.

References

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